

§2-6 Limit at Infinity; Horizontal Asymptotes

Homework : 4,7,17,22,23,29,34,35(c),41

(I) Vertical Asymptotes (垂直漸近線) (V.A.).

(II) Horizontal Asymptotes (水平漸近線) (H.A.).

Definition :

(i) If either of the following 4 limits holds.

$$\lim_{x \rightarrow c^{\pm}} f(x) = \pm\infty,$$

then $x = c$ is called a V.A. of f .

$$(ii) \lim_{x \rightarrow +\infty} f(x) = d \text{ or } \lim_{x \rightarrow -\infty} f(x) = d,$$

then $y = d$ is called a H.A. of f .

Example 1 : $\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 3} = ?$ H.A.=? V.A.=?

Solution : $\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 3} = \lim_{x \rightarrow \infty} \frac{3x^2}{5x^2} = \frac{3}{5}$

$$\text{H.A. } y = \frac{3}{5}$$

$$\text{V.A. No } (\because 5x^2 + 4x + 3 > 0 \text{ for all } x).$$

Example 2 : (i) $f(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}$, (ii) $f(x) = \frac{\sqrt{9x^6 - x}}{x^3 + 1}$ H.A.=? V.A.=?

Solution : (i) H.A.: $y = \frac{\sqrt{2}}{3}$ or $y = -\frac{\sqrt{2}}{3}$

$$\text{V.A.: } x = \frac{5}{3}$$

(ii) H.A.: $y = 3$ or $y = -3$

(iii) V.A.: $x = -1$.

Example 3 : $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) = ?$

Solution :

$$\begin{aligned} & \lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) \\ &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 1} + x)}{(\sqrt{x^2 + 1} + x)} (\sqrt{x^2 + 1} - x) \\ &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2 + 1} + x} \\ &= 0 \end{aligned}$$

註：1. 這個例子 $y = x$ 稱為 $y = \sqrt{x^2 + 1}$ 的 slant (斜) asymptote.

2. 一般來說，若 $f(x) \neq ax + b$ when $|x|$ is “large”. 此處 $a \neq 0$. 且

$$\lim_{x \rightarrow \infty} (f(x) - ax - b) = 0 \text{ 或 } \lim_{x \rightarrow -\infty} (f(x) - ax - b) = 0 \text{ 則 } y = ax + b \text{ 爲}$$

$y = f(x)$ 的斜漸近線。